Reasoning Your Way Out of Anxiety? A Critical Review of Cognitive Restructuring, Executive Function and Adult ADHD

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Abstract

Cognitive behavioral therapy (CBT) is the best-studied psychotherapy for anxiety, yet only about half of treated patients respond, and the mechanism usually offered for its cognitive component, that patients change feelings by examining and replacing automatic thoughts, has seldom been tested against the cognitive demands it places on them. We ask whether cognitive restructuring is an executive-function task that anxiety and attention-deficit/hyperactivity disorder (ADHD) make harder, and whether it is the part of CBT that carries the benefit. We specify the executive demands of restructuring, review evidence on stress, anxiety and prefrontal function (including a check of the popular "prefrontal cortex goes off-line" claim against its sources), review efficacy, bias and component evidence, review alternatives and CBT in adult ADHD, and assess the sparse direct evidence on executive function as a moderator of outcome. About 300 sources were DOI-verified. The evidence supports a narrow claim and not a broad one. Reappraisal draws on working memory in the laboratory, but evidence that stress abolishes it is contested. CBT is effective on average, though effects shrink with active controls and bias adjustment and have not grown over three decades. Component studies, though underpowered, repeatedly fail to find added benefit from restructuring over exposure or behavioural activation. Executive function predicts uptake of the cognitive skill in older adults more than overall outcome. In adult ADHD, effective CBT protocols look behavioural, and

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we found no study of ADHD as a moderator of CBT for anxiety. We close with preregistrable predictions and a research agenda. **Draft 2; for the author's review.**

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1. Introduction

Cognitive behavioral therapy (CBT) is among the most heavily tested psychotherapies. A meta-analysis of 41 placebo-controlled trials in anxiety and related disorders (total $N = 2{,}843$) found that CBT outperformed placebo on target symptoms with a moderate effect (Hedges' $g = 0.56$) (Carpenter et al., 2018). The same literature also shows a ceiling: across 87 studies only about half of patients respond to CBT for anxiety disorders, and the figure depends on how response is defined (Loerinc et al., 2015); a larger meta-analysis of 100 articles reports a mean remission rate of 51.0% (Springer et al., 2018). A therapy can work on average and still leave open *why* it works and *for whom*.

The textbook account of CBT's cognitive component is a top-down loop. A situation triggers an automatic thought ("I am going to fail this talk"); the patient records it, names the distortion, weighs evidence for and against, and generates a more balanced thought; the feeling is expected to follow. This review asks a narrow question about that loop: *does the mechanism ask the patient to use a capacity that anxiety, and in some patients attention-deficit/hyperactivity disorder (ADHD), has already degraded?* If so, the cognitive component should depend on baseline executive capacity, should be harder to use when stressed than when calm, and should contribute less to outcome than behavioural elements that do not demand effortful reasoning.

The argument has a popular version. Judson Brewer, a clinician-researcher, argues that anxiety takes the prefrontal cortex "off-line," so that willpower and reasoning are unavailable at the moments when they are needed (Brewer, 2016, 2021), and he has applied this directly to CBT (Brewer, 2026). Our aim is to find out how much of this survives contact with the primary literature. Our answer, in brief, is that a narrow version survives and the strong version does not.

What we find.. (1) Reappraisal does draw on working memory in laboratory studies, but evidence that acute stress abolishes it is thin and contested.

(2) The direct evidence that executive function moderates CBT outcome is sparse and mostly null for overall outcome, with the clearest signal at the level of the cognitive skill itself. (3) Component studies repeatedly find that adding cognitive restructuring to exposure or behavioural activation does not add measurable benefit, but these studies are underpowered and a minority find a benefit. (4) CBT's headline effects shrink with active controls and bias adjustment; whether they shrink over time is disputed. (5) In adult ADHD, CBT is modestly effective, mostly as an adjunct to medication, and the plausible active ingredient is behavioural compensation; we found no trial testing ADHD as a moderator of CBT for anxiety in adults.

Contributions.. (i) An explicit decomposition of cognitive restructuring into executive demands (Section 3); (ii) an evidence-graded review of the stress-and-prefrontal premise, including a check of the "off-line" claim against its sources (Section 4); (iii) a review of efficacy, bias and component evidence (Sections 6–7); (iv) a review of alternatives and of CBT in adult ADHD (Sections 8–9); (v) the direct moderator evidence (Section 10); (vi) preregistrable predictions (Section 12).

2. Method

This is a critical (theoretical) review, not a PRISMA-complete systematic review. It selects the literatures needed to evaluate one mechanistic argument and states what each can and cannot show. Searches ran in October 2026 across PubMed, Europe PMC, Crossref, publisher pages and the reference lists of key meta-analyses, in four slices: (a) efficacy, bias and head-to-head comparisons; (b) mechanisms and component (dismantling) studies, including exposure and third-wave alternatives; (c) executive function, stress and emotion regulation, including executive function as a moderator of treatment; and (d) ADHD (CBT, medication, mindfulness, executive function, anxiety comorbidity). Priority was given to meta-analyses, umbrella reviews, component trials and moderator studies, and to the strongest critiques and defences of CBT.

Verification.. Every reference was resolved through Crossref (title, authors and year compared with the intended citation). Every quoted number was read from the abstract or full text, not recalled. Roughly 300 candidate sources were verified this way; several candidate references proposed from memory were dropped because the DOI did not resolve or resolved to a

different paper. Sources for which only a partial abstract or a secondary summary was available are flagged in `deep-research/sources/`. Popular sources (books, talks, blogs) are cited only for the claims their authors make.

Limitations of the method.. Single-reviewer screening; no pre-registered protocol; no formal risk-of-bias assessment of included reviews; evidence from depression is used where anxiety evidence on a question is thin, and flagged when so. A PRISMA-style umbrella search is the obvious next step (Section 12).

3. What cognitive restructuring asks of the patient

Consider the standard thought-record exercise. The patient must (a) *notice* an automatic thought while it is occurring, (b) hold the thought, the supporting evidence, the contradicting evidence and a candidate alternative in mind at once, (c) suppress the pull of the automatic thought while evaluating it, and (d) shift from being inside the feeling to observing it. Mapped onto the best-known model of executive function, these are three separable components: updating and holding information (working memory), inhibiting a prepotent response, and shifting between mental sets (Miyake et al., 2000).

Laboratory evidence that reappraisal draws on executive resources.. In event-related-potential studies, high working-memory load eliminated the effect of reappraisal on the late positive potential (Gan et al., 2017; Adamczyk et al., 2022). Working-memory capacity predicted reappraisal success (Schmeichel et al., 2008), and reappraising late in an emotional response, once emotion is already intense, costs self-control resources (Sheppes and Meiran, 2008). People also choose differently as intensity rises: they prefer distraction to reappraisal in high-intensity negative situations and reappraisal in low-intensity ones (Sheppes et al., 2011).

Evidence that it works, and can be trained, when load is low.. Instructed reappraisal reduces negative experience in the laboratory (Hofmann et al., 2009) and can be trained (Denny and Ochsner, 2014). So the claim is not that reappraisal fails; it is that it *costs* executive resources, which matters when those resources are scarce.

Two clarifications.. First, this is a task analysis plus laboratory support, not a treatment finding. Clinicians scaffold the load (worksheets, a therapist who prompts for evidence, graded practice). Second, executive function is not a single resource; the argument concerns working memory, inhibition and shifting, not general intelligence. We return to both in Section 11.

4. Stress, anxiety and the prefrontal executive system

4.1. Mechanistic neuroscience

The most developed account is due to Arnsten. Her review of stress signalling describes the prefrontal cortex (PFC) as the most evolved brain region and also the one most sensitive to the detrimental effects of stress exposure (Arnsten, 2009). The proposed mechanism is an inverted-U in catecholamine signalling: moderate norepinephrine and dopamine levels support prefrontal network firing and working memory, while the high levels of uncontrollable stress weaken them and strengthen amygdala and striatal habit circuitry. A later review summarises the result as stress rapidly flipping the brain from reflective to reflexive control of behaviour (Arnsten, 2015). Acute uncontrollable stress impairs PFC-dependent working memory in monkeys and rodents (Arnsten and Goldman-Rakic, 1998). In humans, a meta-analysis finds that acute stress impairs working memory and cognitive flexibility, with more nuanced effects on inhibition and many moderators (Shields et al., 2016), and imaging suggests a reallocation of resources toward salience networks rather than a switch-off (Hermans et al., 2014).

Where the "off-line" language comes from.. The phrase traces to a hedged line in a 1998 monkey study ("Stress may take the PFC 'off-line"'; Arnsten and Goldman-Rakic 1998), and its author still uses it in quotation marks for *uncontrollable* stress (Arnsten, 2020). Brewer and Roy apply it to worry and cite Arnsten's 2009 and 2015 reviews, which are not studies of worry (Brewer and Roy, 2021); their app-based trial ($n = 65$, treatment-as-usual control) did not test a prefrontal mechanism (Roy et al., 2021). We therefore read "offline" as a popularisation of *impaired, in a graded and dose-dependent way*, mainly under uncontrollable stress.

4.2. Does stress abolish reappraisal in humans?

Evidence for this specific link is contested. The founding study found that participants stressed before learning showed no fear reduction after regulation

training (Raio et al., 2013), and a review summarises that acute stress may impair the persistent inhibition of fear, presumably by altering prefrontal function (Raio and Phelps, 2015). Five later studies ($n = 54$–98) found no group-level impairment (Shermohammed et al., 2017; Sandner et al., 2021; Langer et al., 2021, 2023; Hamza et al., 2024), though two reported sex- or cortisol-linked effects. Their laboratory stressors were moderate, so the null results may not transfer to clinical-level arousal; that gap is itself an open question.

4.3. Cognitive psychology of anxiety

Attentional Control Theory proposes that anxiety impairs the inhibition and shifting functions of executive control, so that anxious individuals are more captured by threat and worse at redirecting attention (Eysenck et al., 2007). The theory predicts that *efficiency* (effort and time) suffers more than *effectiveness* (accuracy), because anxious people compensate with extra effort. Meta-analytic support exists at group level ($g = -0.58$; Shi et al., 2019) and for working memory ($g = -0.334$; Moran, 2016), but with qualifiers: a meta-analysis of 32 studies ($N = 13{,}084$) found generalised anxiety disorder (GAD) associated with poorer cognitive flexibility and working memory but not inhibitory control, and with no association with any executive domain in older adults (Nguyen et al., 2025). Working memory was unrelated to social anxiety in one study (Harrison et al., 2025), and one study found *enhanced* control over negative distractors in GAD (Hallion et al., 2019).

5. Executive function in ADHD

ADHD is heterogeneous: executive difficulties are common but not universal. A meta-analysis of 83 studies found that groups with ADHD underperform controls on every executive task class examined, most strongly on response inhibition, vigilance, working memory and planning, with medium effect sizes (0.46–0.69) (Willcutt et al., 2005). The same authors caution that executive weaknesses are neither necessary nor sufficient to cause all cases of ADHD, and fewer than half of the (mostly child) samples show impairment on any single executive test (Willcutt et al., 2005). Measured more carefully, impairment is more prevalent: in a sample of 136 children, 89% of those with ADHD were impaired on at least one executive function (62% working memory, 27% inhibitory control, 38% set shifting) (Kofler et al., 2019).

Anxiety is common in adult ADHD: a meta-analysis of 58 studies (N = 18,821) found higher anxiety scores in adults with ADHD than without ($g = 0.77$; $g = 0.95$ against non-clinical and $g = 0.44$ against clinical controls) (Hoskins et al., 2026). Whether anxiety adds executive load in ADHD is less clear: in a sample of 339 children, trait anxiety did not uniquely predict working memory or inhibitory control, nor moderate the ADHD effect (Marsh et al., 2024). The stacking assumption (low baseline executive capacity, lowered further by anxiety) is therefore plausible but, in children at least, not supported by the one careful test we found.

6. What the efficacy evidence shows

6.1. The case for CBT

Against placebo, CBT outperforms on anxiety target symptoms ($g = 0.56$ across 41 trials; Carpenter et al. 2018). Against inactive controls, effects are larger: a meta-analysis of 144 trials found $g = 0.75$ (depression) to 0.88 (social anxiety disorder), with 0.80 for GAD and 0.81 for panic disorder (Cuijpers et al., 2016). A 2025 network of 375 trials gave anxiety-spectrum effects between 0.5 and 1.0, above 1.0 for PTSD and specific phobia (Cuijpers et al., 2025). In social anxiety, individual CBT ranked best of the therapies compared in a network meta-analysis of 101 trials (Mayo-Wilson et al., 2014), and in GAD only CBT remained significantly better than treatment as usual at 3–12 months in a network of 65 trials (Papola et al., 2024). Any critique has to explain why CBT works at all.

6.2. The headline shrinks with better controls and bias adjustment

- **Control type.** More than 80% of anxiety trials used waitlist controls, and the few that did not showed much smaller effects (Cuijpers et al., 2016); only 17.4% of trials in that analysis were high quality. In the 375-trial analysis, waitlist-controlled effects exceeded $g = 0.94$, while care-as-usual comparisons gave 0.22–1.13 (Cuijpers et al., 2025).

- **Bias.** In depression, psychotherapy effects fell from $g = 0.70$ to $g = 0.31$ after adjusting for several sources of bias, with only 23% of studies at low risk of bias (Cuijpers et al., 2019a). Of NIH-funded psychotherapy trials, 23.6% never reached publication; including them reduced the pooled effect by about 25% (Driessen et al., 2015). An umbrella review of 102 recent meta-analyses (3,782 RCTs) found most

target-symptom effects small and pooled psychotherapy effects of 0.34 against placebo or treatment as usual (Leichsenring et al., 2022).

- **Response and remission.** About half of anxiety patients respond or remit (Loerinc et al., 2015; Springer et al., 2018).

6.3. Does the effect shrink over time?

This is contested, and the evidence is mixed. In depression, one meta-analysis reported a steady decline in CBT effects since its introduction (Johnsen and Friborg, 2015), but a reanalysis restricted to randomised comparisons judged the decline "most likely a spurious finding" (Cristea et al., 2017), and a 409-trial meta-analysis found no reduction by publication year (Cuijpers et al., 2023). In anxiety, a meta-regression of 49 controlled studies over 30 years found a flat trend (publication year $B = -0.008$, $p = .24$): effects "did not increase over the last 3 decades" (Hofmann et al., 2025). Placebo-controlled anxiety estimates have been reported at lower values in successive reviews (0.73, 0.56, 0.24), but those come from separate reviews with different case mixes, not from one meta-regression. We therefore do *not* claim a decline over time; the defensible statement is that effects have not grown and that they are sensitive to control condition and bias.

7. Where the active ingredients may lie

7.1. Component and dismantling evidence

The foundational dismantling trial (depression, $N = 150$) found no evidence that the complete cognitive treatment outperformed behavioural activation alone or behavioural activation plus automatic-thought skills at termination or at 6 months (Jacobson et al., 1996), and a large non-inferiority trial found behavioural activation as effective as CBT (mean PHQ-9 difference 0.1; Richards et al. 2016). In anxiety:

- In a 108-trial meta-analysis, cognitive therapy, exposure and their combinations were efficacious with "no differential efficacy for any treatment components" (Norton and Price, 2007).

- In a full-factorial trial of internet CBT for social anxiety ($N = 464$), conditions with psychoeducation and exposure were more effective, while conditions with cognitive restructuring or attention training were not superior at post-treatment; the advantage of exposure faded at follow-up (Šipka et al., 2025).

- Adding cognitive restructuring to prolonged exposure did not enhance outcome in assault survivors with PTSD (Foa et al., 2005), nor in children with social anxiety (de Mooij et al., 2023).
- A component network meta-analysis in panic disorder (72 studies) singled out interoceptive exposure, not cognitive restructuring, as efficacy-enhancing (Pompoli et al., 2018).
- In GAD, the network meta-analysis authors concluded that pure cognitive restructuring was not a decisively active ingredient when considered alone (Papola et al., 2024); the paper was later corrected for a duplicated cohort and an error in standard-error handling (JAMA Psychiatry, 2024).

7.2. Evidence against the claim

These findings are not uniform.

- Cognitive therapy beat exposure in social phobia in a meta-analysis of 20 direct comparisons ($N = 1{,}308$), with no difference in PTSD, OCD or panic (Ougrin, 2011).
- In a trauma trial, adding cognitive restructuring to exposure was associated with PTSD rates of 31% versus 63–75% without it at 6 months (Bryant et al., 2008).
- In-session cognitive restructuring correlates with outcome ($r = .35$), though from only four studies and correlationally (Ezawa and Hollon, 2023).
- In a single-session trial ($N = 249$), restructuring before exposure worked about as well as behavioural experiments and better than exposure without explicit processing (Johnco et al., 2025).
- The exposure–cognition split is blurred: exposure may itself work by changing the expectancy of harm (Hofmann, 2008).

7.3. Why the component evidence is weak on both sides

A review of 16 component studies concluded that they lack "the statistical power nor the quality to draw any meaningful conclusion," because none except one could detect an effect smaller than $g = 0.55$ (Cuijpers et al., 2017).

Mediation evidence is correlational: cognitive change tracks symptom change however it is produced (Cristea et al., 2015; Lorenzo-Luaces et al., 2015), cognitive mediation of symptom reduction was limited in a depression cohort (median $r = .06$; Vittengl et al., 2014), and reviewers say no common or specific factor meets causal criteria (Kazdin, 2007; Cuijpers et al., 2019b). A null component result is thus weak evidence that restructuring adds nothing, and a positive correlation is weak evidence that it matters.

7.4. The strongest defence: practice when calm

A defender can answer that patients do not restructure during a panic; they practise when calm so the response becomes automatic. We accept the force of this. It changes the claim, though: if rehearsal during calm periods does the work, the active ingredient is practice and habit formation, and the relevant question is the executive cost of each rehearsal. Exposure, which is central to anxiety CBT, is behavioural, and optimised exposure protocols are explicitly framed around expectancy violation and extinction (Craske et al., 2014).

8. Alternatives and their evidence

8.1. Mindfulness and acceptance-based treatments

The alternative that follows from the argument is to train attention and acceptance directly rather than dispute thoughts. Brewer frames anxiety as a habit loop (trigger, worry, a feeling of control) and proposes noticing, curiosity about the reward value of worry, and a "bigger, better offer" (Brewer, 2021). Evidence for the specific programme is early: in a randomised trial of an app-based programme for GAD ($n = 65$), anxiety scores fell 67% versus 14% under usual care, with mediation implicating reduced worry (Roy et al., 2021). The trial is small, self-reported, compared with usual care rather than CBT, and has the programme's developer among its authors.

Against evidence-based comparators, mindfulness and acceptance approaches show parity, not superiority. A meta-analysis found essentially no difference ($g \approx 0.00$) between mindfulness-based interventions and evidence-based treatments, often CBT (Goldberg et al., 2018); another found no difference from traditional CBT or behavioural therapies ($g = -.07$, nine comparisons; Khoury et al. 2013). For acceptance and commitment therapy (ACT) versus CBT, the pooled difference was non-significant in one meta-analysis (A-Tjak et al., 2014), and a sceptical reading of 60 RCTs concluded that

ACT is not yet well-established for any disorder (Öst, 2014). In 276 adults with anxiety disorders, eight weeks of mindfulness-based stress reduction was noninferior to escitalopram, with fewer discontinuations for adverse events (Hoge et al., 2023). That is parity with a first-line drug, not superiority to CBT. The convergence of camps is itself informative: if CBT, behavioural activation, ACT and mindfulness are roughly equivalent, either common factors dominate or several mechanisms lead to the same place.

8.2. Medication in ADHD

In a network meta-analysis of 133 double-blind trials, amphetamines (standardised effect −0.79) and methylphenidate (−0.49) outperformed placebo in adults on clinician ratings, over short follow-up (Cortese et al., 2018). The certainty varied from high or moderate to low or very low. A Cochrane review of extended-release methylphenidate found a self-rated effect of −0.37, judged 20 of 21 trials at high risk of bias, and noted that all trials excluded participants with psychiatric comorbidity such as anxiety or selected previous responders (Boesen et al., 2022). So the pivotal medication trials say little about adults with ADHD *and* anxiety.

9. CBT in adult ADHD

9.1. Efficacy

A randomised trial of 86 medication-treated adults found that CBT outperformed relaxation with educational support, with 53% versus 23% responders on clinician global ratings (Safren et al., 2010); a smaller earlier trial found 56% versus 13% responders for CBT added to medication (Safren et al., 2005). A meta-analysis of 32 studies found a between-group effect of $g = .65$ on symptoms, smaller with active controls, and neither medication status nor treatment length moderated outcome (Knouse et al., 2017). Another found SMD 0.76 against waitlist but 0.43 against active controls (Young et al., 2020), and a 2026 meta-analysis of 14 RCTs found SMD −0.45 on core symptoms and −0.43 on executive function (Liu et al., 2026). A component network meta-analysis of 113 RCTs found CBT better than placebo only on clinician-reported measures (Ostinelli et al., 2025), and blinded-rater effects of psychological treatments were attenuated to non-significance in an earlier synthesis that did not separate adult from child samples (Sonuga-Barke et al., 2013). An umbrella review judged CBT, atomoxetine and methylphenidate to have at least moderate-certainty evidence in adults, with no high-certainty

long-term evidence for any intervention (Gosling et al., 2025). The Cochrane review rates the CBT evidence as low quality (Lopez et al., 2018).

9.2. What the protocols contain, and where CBT sits

The effective protocols are skills-heavy: time management, organisation and planning, with adaptive thinking as one module (Safren et al., 2010; Solanto et al., 2010). One review argued that learning behavioural compensatory strategies "may be a critical active ingredient" (Knouse and Safren, 2010), and the authors of one trial proposed behavioural control as the key mechanism (Young et al., 2017). We found no dismantling study of cognitive restructuring in adult ADHD, so this is inference. In the largest multi-centre trial ($N = 433$), group CBT did not beat structured clinical management, while methylphenidate beat placebo (Philipsen et al., 2015). Guidelines position CBT as an adjunct to medication (Kooij et al., 2019; National Institute for Health and Care Excellence, 2018). Results on adding medication to CBT are mixed: no incremental benefit in a small double-blind pilot (Weiss et al., 2012), but greater early improvement in an unblinded trial ($N = 88$), with the gap narrowing over follow-up (Cherkasova et al., 2020).

9.3. Anxiety in ADHD

In a meta-analysis of 13 RCTs ($n = 744$), CBT-based interventions reduced anxiety in ADHD modestly (SMD -0.29 post-treatment, -0.42 at follow-up), with effects more apparent in adults; most trials delivered ADHD-focused rather than anxiety-targeted CBT (Wang et al., 2026). Another meta-analysis found CBT improved anxiety, particularly at follow-up, predicted by core-symptom reduction (López-Pinar et al., 2020). In children with primary anxiety in group CBT ($N = 842$), neither ADHD diagnosis nor subtype predicted response or remission, which cuts against the strong form of our thesis (Gould et al., 2018). The adult version of that question appears untested.

10. Does executive function moderate CBT outcome?

This is the question on which the argument stands or falls. The direct evidence is thin and mostly from older adults.

- **Supports (at the level of the skill).** In older adults, poorer cognitive flexibility predicted poorer cognitive-restructuring quality and less

distress reduction from restructuring, but *not* overall CBT outcome: those with poorer flexibility "can still benefit from standardised CBT" (Johnco et al., 2014).

- **Supports (pilot).** In a pilot of CBT for GAD in older adults, participants whose executive scores remained low did not respond, while those whose scores improved responded as well as an intact-executive group (Mohlman and Gorman, 2005).
- **Null or mixed.** No baseline executive effect was found in adult OCD (Vandborg et al., 2016), child anxiety (Godovich et al., 2020) or brief psychotherapy for late-life depression (Schurr et al., 2025); a systematic review of late-life anxiety found no replicated executive predictor (Kyriacou et al., 2023). In a meta-analysis of antidepressant response in late-life depression, only planning and organisation predicted outcome (Pimontel et al., 2016).
- **Opposite direction.** Better executive performance went with non-response to exposure-based CBT in paediatric OCD (Hybel et al., 2017), and worse set-shifting went with better CBT response in late-life depression (Goodkind et al., 2016).
- **Adaptation helps.** Problem-solving therapy beat supportive therapy in older adults with executive impairment, with a larger advantage at greater impairment (Areán et al., 2010), which points to adapting therapy rather than abandoning it.

Taken together, "low executive function means CBT fails" is not supported. A narrower claim is: executive capacity may limit uptake of the *cognitive skill* without limiting overall response, because CBT also works through routes that do not need it. We found no adequately powered study of working-age adults with ADHD and anxiety.

11. The strongest objections

1. **CBT has a large evidence base.** Yes (Section 6). The argument concerns mechanism and moderators, not average efficacy.
2. **CBT for adult ADHD works.** It beats matched controls in medicated adults with a skills-heavy protocol; the evidence is graded low to moderate (Section 9).

3. **Patients practise when calm.** Accepted (Section 7); it relocates the active ingredient to practice.
4. **Exposure is behavioural.** Likewise; it strengthens the argument against restructuring specifically, though exposure may act via changed expectancies (Hofmann, 2008).
5. **Mindfulness is not better.** True. The argument needs only that restructuring is not uniquely necessary.
6. **Cognitive therapy sometimes wins.** Yes, in social phobia (Ougrin, 2011) and in one PTSD trial (Bryant et al., 2008).
7. **The neuroscience is about acute stress.** Correct, and the human reappraisal-under-stress findings are mixed (Section 4).
8. **Executive function does not predict CBT outcome.** Largely true for overall outcome (Section 10); the argument survives only in the narrower, skill-level form.
9. **Therapists scaffold the executive load.** Plausible and untested; a therapist who generates the evidence list lowers demand and could erase a moderator effect.

Limitations. Narrative review by a non-clinician; single-reviewer screening; several key studies are in older adults or children; evidence from depression stands in for anxiety on some questions (component trials, bias adjustment, decline over time).

12. Predictions and a research agenda

The argument yields predictions that could fail.

1. **P1 (skill-level moderation).** Baseline executive function predicts acquisition of restructuring skill more than acquisition of exposure-based skills, and predicts overall outcome only weakly.
2. **P2 (stress-dependence).** The gap between calm and in-the-moment reappraisal is larger in low-executive-function samples and at higher arousal than the moderate stressors used so far.
3. **P3 (component).** In adults with ADHD and anxiety, an exposure-and-behaviour protocol is not inferior to the same protocol plus restructuring.
4. **P4 (ADHD).** ADHD status predicts less benefit from the cognitive module specifically.

What would refute the argument: an adequately powered study finding no moderation of skill acquisition by executive function; a component trial showing that restructuring adds substantial benefit over exposure; or meta-regression showing executive-function-heavy samples respond as well. The cheapest next steps are (i) an individual-participant or meta-regression analysis of existing trials (see `experiments/`), (ii) a PRISMA-style umbrella search to lift this review's method to the target journal's standard, and (iii) a power calculation for a moderator trial.

13. Conclusion

Restructuring-heavy CBT is effective on average, though less so against active controls and after bias adjustment, and its effects have not grown over three decades. What is not established is that the cognitive component is the active ingredient, or that it fails for people with low executive function. The best-supported claims are narrow: reappraisal draws on working memory; component studies, though underpowered, repeatedly fail to find an added benefit of restructuring over behavioural elements; executive capacity may limit the uptake of the cognitive skill but not overall response; and in adult ADHD the effective ingredients look behavioural. The strong, popular form of the argument, that anxiety takes the prefrontal cortex "off-line," is not supported by the primary literature. Direct tests in adults with ADHD and anxiety are the missing evidence.

Declarations

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